Brief Relation Events within Persistent Entity Pairs: A Research and Implementation Draft

Anonymous CVPR submission

Abstract

Video relation detection associates entity pairs with predicates and temporal intervals. Persistent entity tracks provide useful context, yet a brief relationship may occupy only a small part of their co-presence. We study the localization of such occurrences while preserving enduring relations and a fixed prediction budget. Existing project diagnostics identify a substantial residual interval-formation deficit, including failures after correct entity inputs are provided. They do not identify a unique causal mechanism. ***[METHOD: replace with the implemented and novelty-audited mechanism.] [RESULTS: insert only after the matched evaluations are completed.]***

1. Introduction

Video visual relation detection describes who interacts with whom, through which predicate, and during which interval. Entity co-presence and relation occurrence are distinct temporal quantities: two objects can remain visible while a particular interaction appears only briefly. A useful detector must recover that brief occurrence without fragmenting relationships that genuinely persist.

VrdONE jointly predicts predicates and instance masks, while BIG separates classification from multi-instance temporal grounding [1, 2]. SegPVSG explicitly localizes salient segments before recognition [4]. These approaches establish temporal modeling as an existing capability, making a generic claim of "adding temporal localization" insufficient.

Our starting observation comes from a fixed VidOR comparison. At Top100, both BIG and VrdONE miss 25,373 of 30,142 annotated instances. In the subsecond group, 3,151 of 3,167 instances are jointly missed. This paired comparison describes released systems under their existing inputs; it is not an architecture-only controlled comparison. Subsequent fixed-entity diagnostics narrow the problem: on a selected 400-instance persistent-pair short-event cohort, native hull intervals cover 47 instances, while a fixed single-component rule covers 53. Neither result is official predicate-aware relation detection.

We ask how to recover supported brief occurrences under an unchanged prediction budget while retaining enduring relations. The central design must specify why its selected boundaries follow predicate evidence, and its evaluation must distinguish localization improvements from extra candidates or changed scores. **[Finalize contributions after method implementation and evaluation.]**

2. Related Work

Video relation modeling. BIG formulates classification and temporal grounding through a bipartite graph, and VrdONE predicts relation instances through one-dimensional instance segmentation [1, 2]. SegPVSG addresses temporally sparse interactions through temporal focusing and relation-centric video augmentation [4]. Our comparison must therefore isolate a more specific contribution than sparse event localization alone.

Temporal event localization. VSGN directly targets short actions using temporal magnification and cross-scale graphs [9]. ActionFormer and TriDet offer strong multi-scale temporal detection and boundary modeling designs [5, 7]. BAM-DETR uses boundary-oriented decoding, while DenoiseLoc learns interval refinement through denoising [3, 6]. These methods motivate explicit controls for boundary prediction, refinement, and localization quality.

Conditional span prediction. VSLNet uses query-guided highlighting and span prediction [8]. Consequently, a conditioned start/end head alone is a close-precedent baseline. **[Compare the final core computation against the closest verified methods.]**

3. Problem and Empirical Basis

For an ordered entity pair, let $C = [0, T)$ denote a continuous co-presence interval, and $X = \{x_t\}$ its observed features. A relation event has predicate p and interval $I^* = [s^*, e^*)$. Its temporal occupancy is

$$\pi = \frac{e^* - s^*}{T}. \tag{1}$$

Table 1. Historical paired Top100 outcomes on 30,142 VidOR GT instances. These are observations of two existing systems, not matched-input architecture ablations.

Outcome	Count / total
Both detected	2,487 / 30,142
VrdONE only	1,809 / 30,142
Neither detected	25,373 / 30,142
BIG only	473 / 30,142

The detector predicts a set of predicate-labeled intervals, allowing simultaneous predicates and multiple separated occurrences of one predicate. Inference does not receive GT predicates, durations, or event boundaries.

For temporal diagnosis, $\mathrm{tIoU}(I, I^*) = |I \cap I^*|/|I \cup I^*|$. If a prediction contains the entire event but adds duration Δ, then

$$\mathrm{tIoU} = \frac{L}{L + \Delta}, \qquad L = |I^*|. \tag{2}$$

This establishes sensitivity to absolute boundary error for short events; it does not establish why a particular model makes that error. Official relation detection additionally requires the correct triplet and both entity tube matches.

The fixed 400-instance diagnosis uses already-exposed videos and correct-entity inputs. Its 47-to-53 interval coverage change is a limited readout result, not an end-to-end method gain. The independent-query upper bounds and oracle choices must remain separate from deployable inference.

4. Method

[DESIGN ONLY: this section must be replaced by the final implemented method.]

The current candidate design question is how a predicate-conditioned interval can explain the observed evidence without including unsupported co-presence or discarding valid event evidence. One abstract formulation is

$$S(p, I \mid X) = S_{\text{support}} - \lambda S_{\text{extension}} - \mu S_{\text{omission}}. \tag{3}$$

This equation is a design requirement, not an implemented algorithm or a novelty claim. The completed method must define all three terms, their normalization, training targets, inference computation, and treatment of repeated occurrences.

Required architecture specification. **[Define tensor shapes, predicate conditioning, query/event assignment, legal coordinates, interval selection, confidence, and fixed-budget inference.]**

Table 2. Planned official detection evaluation. New evaluations are not run. Historical diagnostic numbers are deliberately not substituted here.

Method	mAP	R@50	R@100
Frozen base	**[NOT RUN]**	**[NOT RUN]**	**[NOT RUN]**
Native FT	**[NOT RUN]**	**[NOT RUN]**	**[NOT RUN]**
Boundary	**[NOT RUN]**	**[NOT RUN]**	**[NOT RUN]**
TFN-style	**[NOT RUN]**	**[NOT RUN]**	**[NOT RUN]**
Proposed	**[NOT RUN]**	**[NOT RUN]**	**[NOT RUN]**

Training objective. **[Write only losses used in code. Specify native initialization, frozen parameters, matching, unknown-label handling, and train-only statistics.]**

Ordinary boundary control. A start/end baseline can select legal endpoints using

$$\arg\max_{i<j} \left[\log p_s(i) + \log p_e(j) + \lambda_L \log p(L = j - i \mid p)\right]. \tag{4}$$

Conditioned boundaries and duration priors have close precedent. Their inclusion does not itself establish the main contribution, and true long events must retain support throughout the duration prior.

5. Experiments

Data and protocols. VidOR is the primary benchmark. Official train, development, and evaluation video IDs must be sealed before new experiments. The previously examined 27- and 250-video diagnostic subsets are not independent held-out tests. ImageNet-VidVRD can provide an additional benchmark, with its different annotation granularity reported explicitly. A PVSG comparison requires task adaptation and a VidOR source-video overlap audit.

Comparators. Compare the unchanged checkpoint, same-budget native continued training, an ordinary boundary control, a clearly documented temporal-focusing adaptation, and the proposed method. Use the same inputs, budget, evaluation implementation, and matched seeds. A task-adapted control is not automatically an author-method reproduction.

Event slices and gains/losses. Report < 1, 1–3, 3–10, and ≥ 10 second groups, continuous co-presence, interval occupancy, and both predicate-aware and time-only results. Every recovery analysis includes newly detected instances, lost baseline instances, fixed-original-C recovery, output counts, and long-event protection. Ratios include their actual numerator and denominator; AP is not a TP/GT ratio.

Mechanism and ablations. **[Pre-register the final core mechanism comparator and necessary ablations. Do not repeat failed recipes under new names or choose validation thresholds after seeing their results.]**

Reproducibility. **[Add actual environments, weights, splits, trainable parameters, seeds, runtime, uncertainty, prediction seals, and code commit after execution.]**

6. Limitations and Conclusion

The existing project evidence localizes a persistent short-interval deficit but does not identify a unique causal explanation. Some event boundaries, especially gaze relations, are also visually ambiguous. Frame-rate and annotation differences limit direct cross-dataset comparisons.

[After implementation and evaluation, conclude only the supported method and mechanism findings. Do not claim improved performance or a complete causal account from this scaffold.]

References

[1] Kaifeng Gao, Long Chen, Yulei Niu, Jian Shao, and Jun Xiao. Classification-then-grounding: Reformulating video scene graphs as temporal bipartite graphs. In *CVPR*, 2022. 1

[2] Xinjie Jiang, Chenxi Zheng, Xuemiao Xu, Bangzhen Liu, Weiying Zheng, Huaidong Zhang, and Shengfeng He. Vrdone: One-stage video visual relation detection. arXiv:2408.09408, 2024. 1

[3] Pilhyeon Lee and Hyeran Byun. Bam-detr: Boundary-aligned moment detection transformer for temporal sentence grounding in videos. In *ECCV*, 2024. 1

[4] Yikai Li, Quhui Ke, Jinglin Liang, Zhi-Yuan Zhang, Zhidi Lin, and Shuangping Huang. Segpvsg: Panoptic video scene graph generation via temporal focusing and generative augmentation. In *ICML*, pages 67742–67759, 2026. 1

[5] Dingfeng Shi, Yujie Zhong, Qiong Cao, Lin Ma, Jia Li, and Dacheng Tao. Tridet: Temporal action detection with relative boundary modeling. In *CVPR*, 2023. 1

[6] Mengmeng Xu, Mattia Soldan, Jialin Gao, Shuming Liu, Juan-Manuel Pérez-Rúa, and Bernard Ghanem. Boundary-denoising for video activity localization. arXiv:2304.02934, 2023. 1

[7] Chenlin Zhang, Jianxin Wu, and Yin Li. Actionformer: Localizing moments of actions with transformers. In *ECCV*, 2022. 1

[8] Hao Zhang, Aixin Sun, Wei Jing, and Joey Tianyi Zhou. Span-based localizing network for natural language video localization. In *ACL*, 2020. 1

[9] Chen Zhao, Ali Thabet, and Bernard Ghanem. Video self-stitching graph network for temporal action localization. In *ICCV*, 2021. 1

LLM Usage Statement

Language models assisted with organization, drafting, and preparation of this research scaffold. Authors must update this statement to reflect the actual design, implementation, writing, and verification performed for the completed paper.